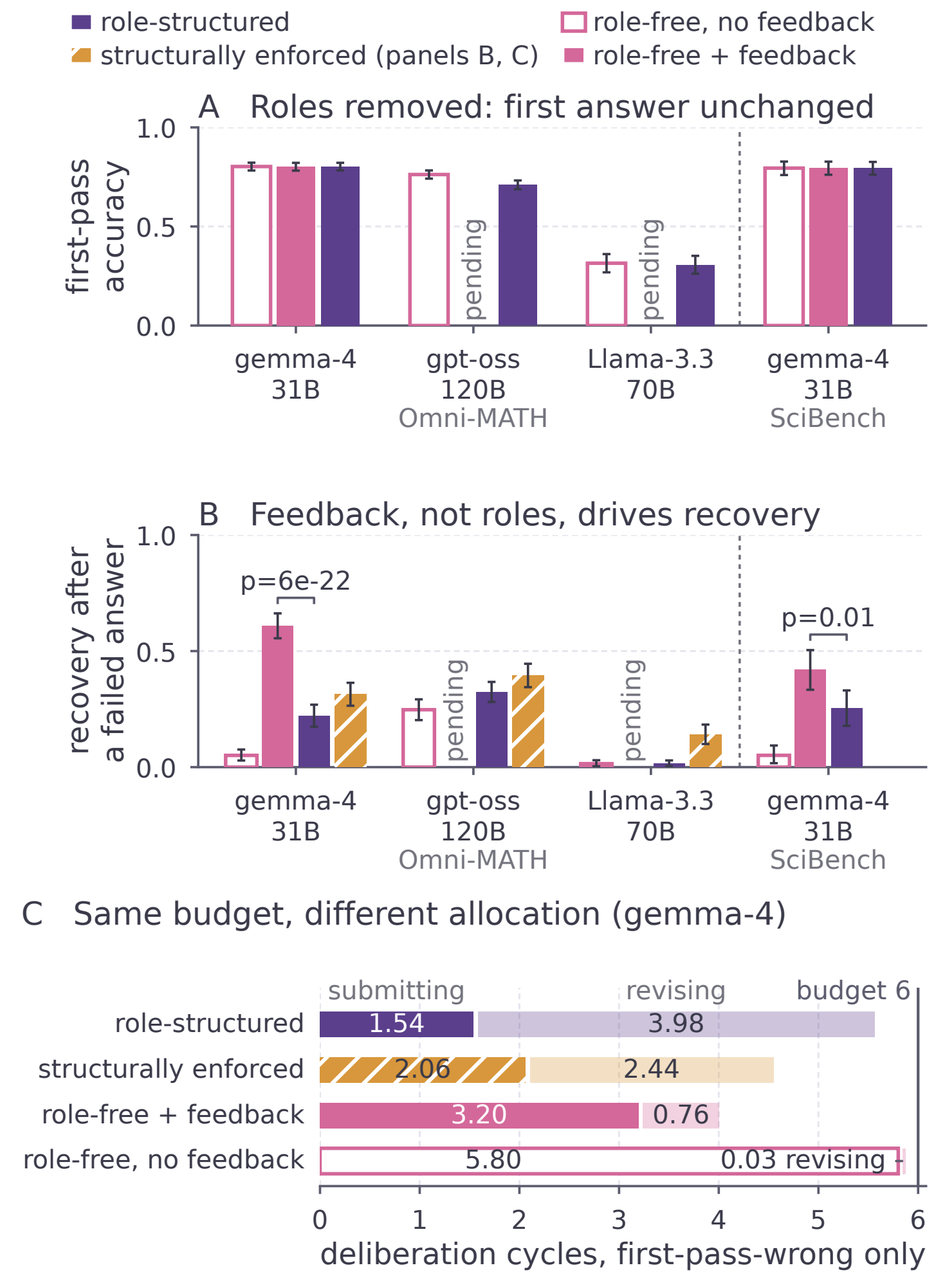
role-structured
structurally enforced (panels B, C)
role-free, no feedback
role-free + feedback
A Roles removed: first answer unchanged
first-pass accuracy
1.0
0.5
0.0
pending
pending
gemma-4 31B
gpt-oss 120B
Omni-MATH
Llama-3.3 70B
gemma-4 31B
SciBench
B Feedback, not roles, drives recovery
recovery after a failed answer
p=6e-22
p=0.01
C Same budget, different allocation (gemma-4)
submitting
revising
budget 6
role-structured
1.54
3.98
structurally enforced
2.06
2.44
role-free + feedback
3.20
0.76
role-free, no feedback
5.80
0.03 revising →
0
1
2
3
4
5
6
deliberation cycles, first-pass-wrong only